

Tissue-engineered blood vessels for clinical translation: Design logic, representative advances and persistent barriers

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Abstract

Tissue-engineered blood vessels are promising alternatives for small-diameter vascular reconstruction, but clinical translation remains limited by thrombosis, compliance mismatch, delayed endothelialization and unstable host remodeling. This review synthesizes representative advances in scaffold design, seed-cell selection, bioactive regulation and bio-fabrication, emphasizing how these components interact under hemodynamic load. Rather than cataloguing technologies, we argue that translational success depends on matching material architecture, cellular phenotype, immune remodeling and manufacturing constraints to specific clinical indications. Acellular or host-repopulating grafts may be most realistic for urgent trauma or vascular access, whereas coronary and distal peripheral reconstruction require tighter control of endothelialization and mechanics. Future tissue-engineered vessels should be evaluated as indication-specific products with clinically meaningful benchmarks beyond short-term patency.

Keywords

tissue-engineered blood vessels, vascular grafts, tissue engineering, endothelialization, biomaterials, biofabrication, vascular regeneration

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1. Introduction

Blood vessel substitutes are routinely used in cardiovascular and vascular surgery, but the unmet need is concentrated in small-diameter conduits, where thrombosis, compliance mismatch and maladaptive remodeling remain difficult to control. In this setting, the question is no longer whether an artificial conduit can be implanted, but whether it can survive the sequential demands of early hemocompatibility, intermediate remodeling and long-term biological integration.^{1–3}

Current clinical options highlight this gap. Autologous vein or artery remains the reference standard, but availability, harvest morbidity, prior vascular disease and caliber mismatch limit use in many patients.^{4,5} Permanent synthetic grafts perform acceptably in large-caliber indications, yet their performance declines sharply in small-caliber reconstruction

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because non-endothelialized and mechanically mismatched luminal surfaces are exposed to pulsatile blood flow during the period of greatest thrombotic vulnerability.^{6–8}

Biodegradable polymers and composite scaffolds were introduced to overcome the inertness of permanent prostheses, but this shift did not solve the problem by itself. A scaffold that degrades too slowly can prolong the foreign-body burden and impair cell ingress; one that degrades too quickly can lose structural support before neo-tissue matures.^{9–17} This temporal mismatch between material resorption and host regeneration remains one of the most important reasons why encouraging preclinical results do not consistently translate into durable vascular grafts.

TEBVs therefore should not be evaluated as simple tubes assembled from biomaterials and cells, but as dynamic systems in which scaffold architecture, endothelialization, immune response, cellular phenotype and manufacturing route interact under hemodynamic load.^{18–21} In practice, translational success depends on whether these components are coordinated rather than maximized individually.

In this review, the term tissue-engineered blood vessel (TEBV) is used broadly to include tissue-engineered vascular grafts (TEVGs) and related bioengineered vascular conduits, unless a cited study uses a more specific term. This definition is intended to keep the discussion clinically inclusive while preserving the distinction between scaffold-based, cell-seeded, acellular and host-repopulating strategies.

This Review preserves a comprehensive synthesis while shifting the emphasis from cataloguing technologies to judging their translational value. We discuss representative studies in biomaterials, seed-cell selection, auxiliary bioactive systems, and advanced biofabrication, and we ask three recurring questions: what enables early patency, what supports adaptive remodeling, and what limits durable clinical deployment.

Our central argument is that no universal TEBV is likely to serve all indications. The most credible path forward is indication-specific design, supported by selective bioactivity, realistic manufacturing, and evidence that extends beyond proof-of-concept animal studies. [Figure 1](#) summarizes this design logic.

A clinically useful synthesis must therefore separate what is merely technically feasible from what is clinically credible. For small-diameter grafts, the decisive interval is the early post-implant window, when blood-contacting surfaces are exposed to thrombosis, compliance mismatch amplifies disturbed flow, and inflammatory cues influence whether endothelial coverage, smooth-muscle organization and extracellular-matrix deposition proceed toward regeneration or toward stenosis and calcification. Recent field-wide reviews have converged on this point: the bottleneck is not the absence of elegant platforms, but the scarcity of constructs that remain coherent across mechanics, hemocompatibility, cellular guidance and manufacturability.^{22–24}

This distinction provides the conceptual frame for judging representative studies throughout this review. Rather than asking which platform is universally superior, we evaluate whether each material, cellular or bioactive strategy solves a

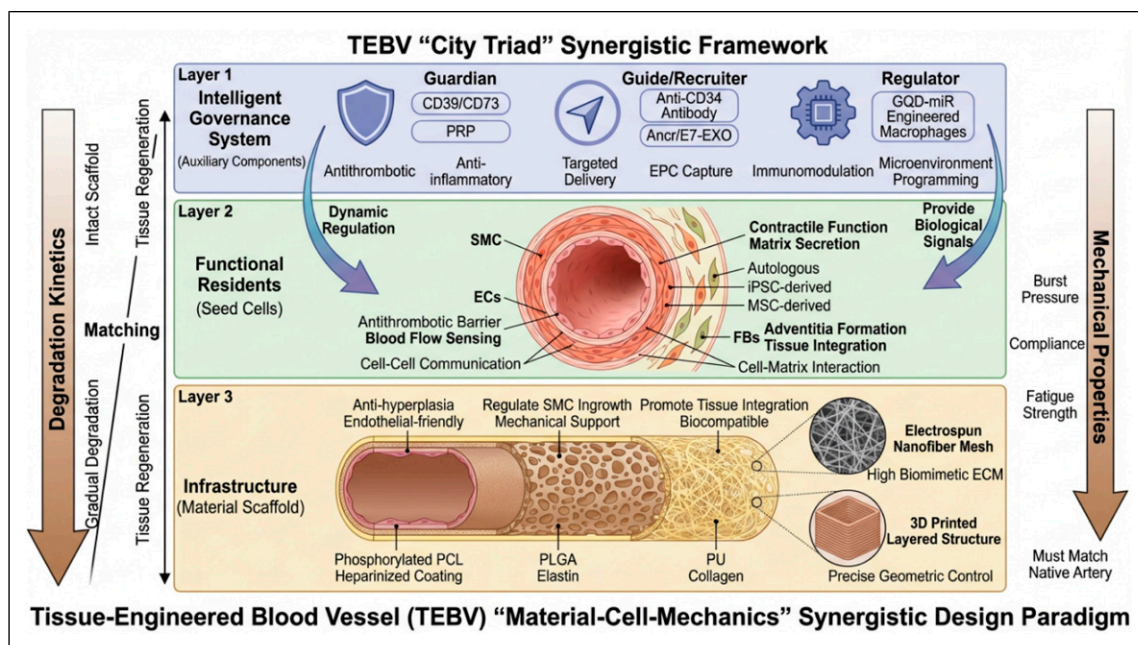


Figure 1. Design framework linking scaffold, cells and bioactive regulators to mechanics, endothelialization and host remodeling in tissue-engineered blood vessels. Created by the authors.

defined translational bottleneck in a defined clinical setting. In this way, the review shifts from platform ranking to evidence-based interpretation of what each design choice can realistically contribute.^{5,8,25–30}

Accordingly, we preserve a comprehensive structure but apply a stricter evidentiary logic. Each section begins with an integrative judgment, then uses a limited number of representative studies to illustrate where the field has genuinely advanced, and finally closes by identifying the unresolved constraint that still prevents routine clinical use. This selective approach is more useful than exhaustive listing because the translational question is no longer whether tissue-engineered vessels can be built; it is which combinations of materials, cells, adjunctive biology and fabrication strategy can be justified for a defined clinical task.^{24,31–33}

In Figure 1, the terms “guardian,” “guide-recruiter” and “regulator” are used only as shorthand labels for three auxiliary functions: early antithrombotic and anti-inflammatory protection, directed endothelial or progenitor-cell recruitment, and longer-term immune or remodeling regulation. These terms are not intended to define independent material classes; rather, they help organize the bioactive adjuncts discussed in Section 2.3.

2. Design determinants of tissue-engineered blood vessels

2.1. Material innovation and degradation kinetics

The key question in TEBV design is not which individual component appears most sophisticated, but whether the combined construct can remain patent during the earliest thrombotic window and then remodel without losing structural integrity. The following sections review representative material, cellular and microenvironmental strategies through that translational lens.

2.1.1. Synthetic materials: Balancing mechanical integrity and bioactivity

Non-degradable synthetic polymers, especially ePTFE and Dacron, remain instructive benchmarks because they demonstrate the mechanical requirements that any arterial substitute must meet. However, their poor endothelialization and compliance mismatch largely explain why success in large-caliber bypass grafting has not translated to small-diameter applications.^{6–8}

Biodegradable synthetic elastomers such as PLA, PCL and PU were introduced to address this limitation by enabling temporary mechanical support with subsequent host replacement. Their principal advantage is tunability, but their performance is constrained by hydrophobicity, solvent-processing issues and, in some systems, prolonged degradation that delays cell infiltration and can contribute to late calcification or chronic inflammation.^{9–12,14,16,29} Representative work with modified PCL and PU scaffolds shows that synthetic platforms can be improved substantially, but usually only when structural design is paired with surface or biochemical functionalization.^{10–12,14,16}

Hemodynamic evidence reinforces this point. Reviews focused specifically on vascular graft failure have emphasized that wall shear stress, transmural loading and compliance mismatch are not secondary variables but direct regulators of intimal hyperplasia, thrombus formation and wall remodeling.²³ In practice, a polymer that looks promising in static culture may still fail if it creates disturbed anastomotic flow or a stiffness discontinuity that the native artery cannot accommodate. This is one reason why topography, luminal chemistry and wall architecture increasingly matter as much as bulk polymer identity itself: they determine whether the graft behaves as a mechanically tolerated vascular interface rather than as a persistent foreign tube.²³

2.1.2. Natural polymeric materials: Salvaging mechanical fragility

Natural polymeric materials—including chitosan (CS), gelatin (Gel), and sodium alginate (SA)—provide an intrinsically bioactive niche that powerfully mimics the native extracellular matrix (ECM).^{34,35} Gelatin, for instance, is replete with RGD (Arg-Gly-Asp) sequences that naturally facilitate endothelial adhesion and proliferation.^{36–38} However, the clinical translation of these hydrogels faces a critical barrier: they inherently suffer from profound mechanical fragility and unpredictable, rapid degradation rates in vivo, rendering them entirely unviable as standalone load-bearing arterial conduits.

To salvage their structural integrity, aggressive chemical crosslinking (such as EDC/NHS for gelatin or covalent bonding for alginate)^{39–44} or composite blending is mandatory. Composite strategies, including gradient CS/PCL scaffolds and alginate- or gelatin-based hydrogel systems,⁴⁵ illustrate how natural-polymer bioactivity can be combined with synthetic or ionic networks, to improve endothelial behavior and handling properties while compensating for intrinsic mechanical weakness.^{46,47} Moving forward, defining the exact structure-function relationships of novel crosslinking techniques will be essential for their deployment in complex vascular engineering.

Recent reviews of extracellular-matrix-containing vascular biomaterials have strengthened this interpretation. ECM-derived or ECM-inspired molecules can simultaneously influence platelet behavior, endothelial recruitment and smooth-muscle regulation, which is attractive because these three processes are tightly linked in graft failure.²² However, the same

reviews also stress that biological richness does not eliminate the need for manufacturing discipline. Batch effects, degradation variability and sterilization sensitivity remain major barriers, so natural-polymer approaches are most persuasive when they are used selectively – as luminal coatings, interpenetrating networks or compartmentalized bioactive layers – rather than as mechanically dominant standalone conduits.²²

2.1.3. Decellularized matrices

Decellularized matrices are appealing because they preserve native extracellular-matrix architecture and biological cues while reducing cellular immunogenicity.^{48–52} They therefore occupy an important middle ground between fully synthetic scaffolds and fully living grafts.

Their limitation is not conceptual but practical. Effective decellularization can weaken mechanics, expose thrombogenic collagen, and introduce batch variability that complicates standardization. As a result, decellularized grafts often still require secondary antithrombotic treatment, re-endothelialization or other surface modification before they become credible off-the-shelf candidates.^{48–56}

This is precisely why decellularized and host-repopulating grafts continue to attract translational interest despite their imperfections. They offer a plausible middle position between fully living constructs, which are difficult to standardize, and purely synthetic grafts, which often lack instructive biology. The translational question is therefore not whether decellularized matrices perfectly mimic native vessels – they do not – but whether they provide enough biological guidance to justify the remaining manufacturing burden. In current practice, their strongest case lies in applications where host repopulation and infection resistance are valued, and where moderate heterogeneity can be managed through process control rather than ignored.^{22,31,48–52}

2.1.4. Composite scaffold materials: Orchestrating degradation kinetics

Composite scaffolds have emerged because no single material class simultaneously offers native-like mechanics, controlled degradation, manufacturability and instructive bioactivity. Layered or blended systems can distribute functions across components, as illustrated by tri-layer designs that separately prioritize luminal hemocompatibility, medial cell ingress and outer-wall integrity.^{57–59}

The advantage of composite design is therefore not complexity for its own sake, but the ability to synchronize failure modes. A successful composite graft must maintain shape during the early postoperative phase, permit timely cellular repopulation, and relinquish support only when host extracellular matrix has become mechanically meaningful. This coordination of degradation and remodeling, rather than any single material label, is the real design target.

Composite design also better matches how clinical failure actually unfolds over time. Early thrombosis is predominantly a luminal problem, mid-term intimal hyperplasia is often a mechano-biological interface problem, and late dilation or calcification reflects the long-tail interaction between degradation, inflammation and matrix turnover. A layered or hybrid scaffold can in principle assign different tasks to different compartments – for example, an antithrombotic inner layer, a porous cell-instructive medial layer and a mechanically stabilizing outer layer – without forcing one material to do everything at once.^{57–59} This logic is conceptually stronger than searching for a single miracle polymer, but it remains credible only if the resulting architecture can be manufactured reproducibly and validated under realistic hemodynamic loading.^{23,33}

2.2. Cell sources and phenotypic stability

Because native vessels are multicellular organs rather than inert conduits, cell sourcing in TEBV research has always been tied to a dual ambition: to restore endothelial and medial function, and to do so without creating new safety or manufacturing barriers. The most informative comparison is therefore not between ‘primary’ and ‘stem’ cells in the abstract, but between mature function, scalability and phenotypic stability under flow.⁶⁰

2.2.1. Autologous and stem-cell sources: Benchmarks and bottlenecks

Autologous vascular cells remain the biological benchmark. They provide the closest physiological match and avoid allogeneic immune concerns, and landmark scaffold-free approaches have shown that such cells can produce mechanically robust conduits.^{61,62} Their main limitation, however, is the clinical population that needs TEBVs most: patients with advanced vascular disease often have limited, poor-quality or anatomically unsuitable donor tissue.

Stem-cell approaches were developed to address this shortage. ESCs and MSCs each provide useful proof-of-principle – ESCs for vascular differentiation potential, MSCs for accessibility and paracrine support – yet both bring well-known constraints, including ethical concerns, donor-related variability, limited lineage control or reduced potency in aged and diseased hosts.^{63–69} These features do not make stem cells unusable; they simply mean that translation depends on tighter control of maturation and quality than many proof-of-concept studies require.

That translational distinction is becoming sharper as the field matures. The issue is no longer whether stem-cell-derived vascular cells can be produced, but whether they can be produced with batch consistency, lineage purity and release criteria compatible with clinical manufacturing. Reviews published over the past two years increasingly frame cell sourcing as a product-development problem rather than a biology problem alone.^{24,31,33} In other words, a cell source that performs well in a mechanistic study but requires prolonged expansion, expensive quality control or unstable differentiation may still be inferior to a less versatile but more reproducible alternative.

2.2.2. Induced pluripotent stem cells and endothelial progenitor cells: Promise and phenotypic uncertainty

Induced pluripotent stem cells are especially attractive because they combine patient specificity with broad differentiation capacity, and several studies have shown that iPSC-derived vascular cells can support functional tissue engineering.^{70–73} Yet their central translational challenge is not merely tumorigenicity, which has been reduced by non-integrating approaches, but the possibility that incompletely matured derivatives may drift under hemodynamic or inflammatory stress.

Endothelial progenitor cells address a different problem: they are not a universal cell source, but they can accelerate endothelialization and improve early patency, especially when combined with supportive stromal populations.^{74–78} Their value is therefore tactical rather than comprehensive. Across all cell classes, the next step is a shift from source discovery to release criteria: purity, functional phenotype, durability under flow and reproducibility of manufacture.

A clinically useful synthesis should therefore resist ranking cell sources in the abstract. Coronary and distal peripheral grafting demand exceptionally reliable endothelial performance under arterial shear, which raises the bar for phenotypic maturity. Hemodialysis access may benefit more from infection resistance, wall durability and host repopulation than from extensive ex vivo cellular complexity. Trauma repair, by contrast, usually privileges product availability over cell personalization. Once judged through this clinical lens, autologous cells, iPSCs, EPC capture and acellular host-repopulating constructs become complementary strategies rather than competing ideological camps.^{5,25–28,30–33}

2.3. Auxiliary bioactive systems as stage-specific adjuncts

Auxiliary bioactive systems are most useful when they reduce specific biological bottlenecks that a scaffold or cell source alone cannot solve. In the tissue-engineered vascular graft setting, the most important targets are early thrombosis, excessive inflammation, delayed endothelialization and late calcific or fibrotic remodeling.

These systems should be interpreted as stage-specific adjuncts rather than mature stand-alone solutions. Most are supported by in vitro studies, small-animal implantation, coating studies or short-term proof-of-concept experiments. Their translational value therefore depends on whether they address a defined bottleneck without adding unacceptable manufacturing, safety or regulatory complexity. This evidence-aware framing is important because the maturity of CD39/CD73-inspired purinergic regulation, platelet-rich plasma, engineered exosomes, cell-capture coatings and macrophage-based delivery systems is not equivalent to that of established scaffold material classes.

2.3.1. CD39/CD73-inspired purinergic regulation: Antithrombotic and anti-inflammatory logic

CD39/CD73-inspired purinergic antithrombotic strategies represent a shift from passive hemocompatible surfaces toward metabolically active interfaces. By converting ATP/ADP into adenosine, these systems are designed to reduce platelet activation while tempering inflammatory signaling.^{79–81} Its appeal lies in this dual function; its limitation remains the long-term stability and biocompatibility of the enzyme-carrier construct under implantation conditions (Figures 2–4).

In parallel with enzyme-inspired purinergic regulation, marine sulfated polysaccharides have emerged as potential biofunctional alternatives or complements to conventional heparin-based vascular graft functionalization. Fucoidans, carrageenans and fucosylated chondroitin sulfates have attracted interest because their sulfated glycan structures can mimic aspects of the endothelial glycocalyx and may modulate coagulation, inflammation and vascular cell behavior.⁹² Recent scaffold-oriented work has further suggested that marine sulfated polysaccharides can be integrated into multilayered, bioresorbable small-caliber vascular graft designs that combine electrospinning and advanced printing strategies.⁹³ These approaches are promising because they diversify antithrombotic surface chemistry beyond conventional heparinization, but their translation still requires careful control of molecular weight, sulfation pattern, batch variability, anticoagulant potency, endothelial compatibility and long-term stability under flow. Accordingly, marine-derived polysaccharides should be discussed as an emerging functionalization strategy rather than as a clinically mature substitute for established antithrombotic coatings.

PRP illustrates both the promise and the unpredictability of autologous adjuncts. It can temporarily enrich the graft microenvironment with matrix-forming and angiogenic cues, and it may improve the biological performance of otherwise inert scaffolds.^{82–84} Yet its effects are dose-dependent and context-dependent, which makes standardization essential if PRP is to move from a helpful supplement to a reproducible component of graft design.

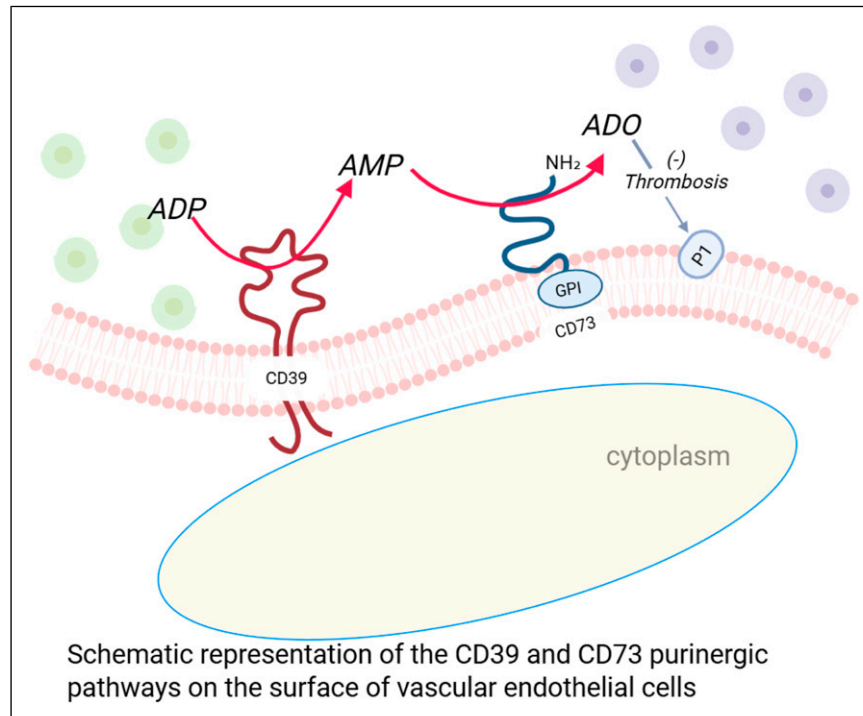


Figure 2. Purine-metabolizing antithrombotic concept for CD39/CD73-based vascular interfaces. Created by the authors based on published mechanistic concepts.^{80,81}

2.3.2. *Ancr/E7-EXO: Epigenetic reprogramming of the pathological niche*

Engineered exosomes such as Ancr/E7-EXO highlight a more targeted strategy: correcting pathological cell fate decisions that drive graft calcification or maladaptive remodeling. In this case, the importance of the platform lies less in exosomes per se than in the demonstration that lineage control within the graft wall can be modulated with molecular precision.^{85–87} Manufacturing scale, target specificity and regulatory complexity, however, remain substantial barriers.

2.3.3. *Anti-CD34 functionalized multilayers: Targeted progenitor-cell capture*

Cell-capture coatings represent a complementary approach aimed at shortening the vulnerable interval before an endothelial barrier is established. Anti-CD34-functionalized heparin-collagen multilayers are representative of this concept: the coating couples anticoagulant activity with recruitment of circulating progenitor cells, thereby attempting to accelerate in situ endothelialization rather than relying solely on pre-seeded cells.^{88,89} The promise is practical simplicity; the limitation is that circulating progenitor pools and capture efficiency vary markedly across patients.

2.3.4. *GQD-miR macrophage bioreactor: Immune programming as a design variable*

Macrophage-based delivery systems push the field toward active immune programming. Representative macrophage-delivery studies suggest that engineered macrophage behavior can shift inflammatory microenvironments toward pro-repair phenotypes and thereby support angiogenesis and graft patency.⁹⁰ These strategies are conceptually important because they treat inflammation as a design variable, but their translational complexity is higher than that of surface or acellular approaches.

This point is worth emphasizing because adjunctive biology can easily become performative. Adding exosomes, platelet products, antibodies or immune-cell engineering is justified only when the adjunct solves a defined bottleneck that scaffold architecture alone cannot solve. The strongest adjunctive strategies are therefore not the most complex ones, but those that shorten the endothelialization lag, suppress pathological inflammation without blunting remodeling, or prevent lineage drift under high-risk conditions such as diabetes or chronic kidney disease.^{82–90} Once the adjunct becomes impossible to scale, too variable to release as a product, or too indication-agnostic to guide patient selection, its translational value declines sharply.

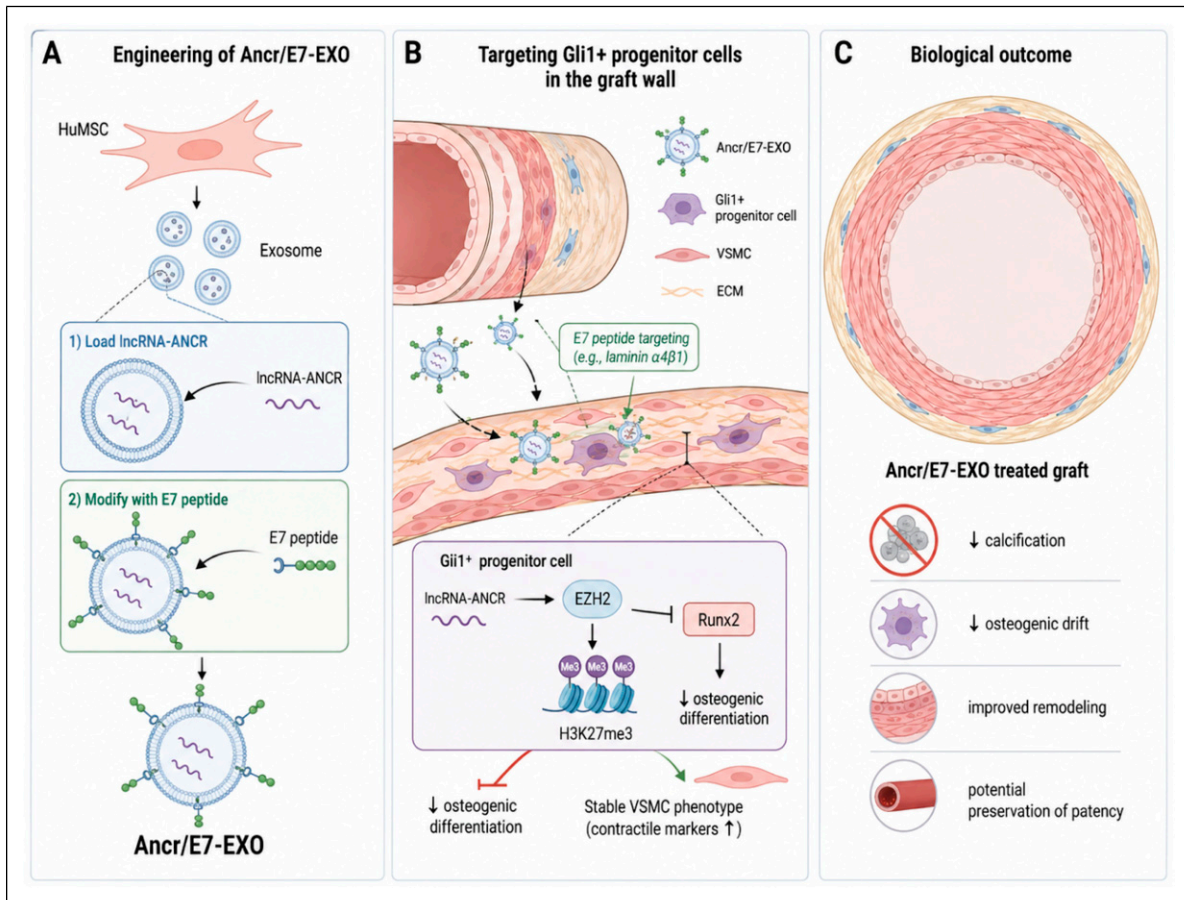


Figure 3. Original schematic of the Ancr/E7-EXO strategy for targeting Gli1-positive progenitor cells and suppressing vascular graft calcification. Created by the authors.

3. Biofabrication and maturation of tissue-engineered vascular grafts

Biofabrication does not simply determine graft shape; it defines pore architecture, fiber alignment, wall heterogeneity, handling properties and the conditions under which cells experience flow and strain. As a result, manufacturing methods should be compared not only by resolution or novelty, but by the specific compromises they impose on biomechanics, biology and scalability.

Recent biomechanical analyses strengthen the argument that fabrication should be judged by the vascular loading environment it creates, not just by printing resolution or fiber diameter.²³ Compliance, anisotropy, kink resistance, suture retention and pulsatile fatigue are not interchangeable surrogates; together they determine whether the host artery experiences an abrupt material discontinuity at the anastomosis and whether luminal flow remains physiologically organized. A sophisticated fabrication method that cannot reproducibly meet these mechanical targets may still fail clinically even if its cell biology is attractive.^{23,31}

3.1. Biological self-assembly and in vivo bioreactors: Biomimicry versus scalability

Self-assembly and in vivo bioreactor strategies occupy the biomimetic end of the spectrum. By minimizing or eliminating exogenous synthetic material, they can generate highly biological conduits with strong matrix composition and, in selected settings, impressive burst strength.^{28,30,61,94–97}

The trade-off is time and predictability. Long culture periods, patient dependence and challenges in scaling production have limited these approaches mainly to niche or proof-of-principle use. Their enduring importance is conceptual: they demonstrate what biological fidelity can achieve, but also why fidelity alone is not the same as clinical feasibility.

Even so, these biologically rich approaches continue to inform the field because they clarify what a mature vascular wall should look like. They also highlight the role of preconditioning. Pulsatile bioreactors, mechanical strain and controlled

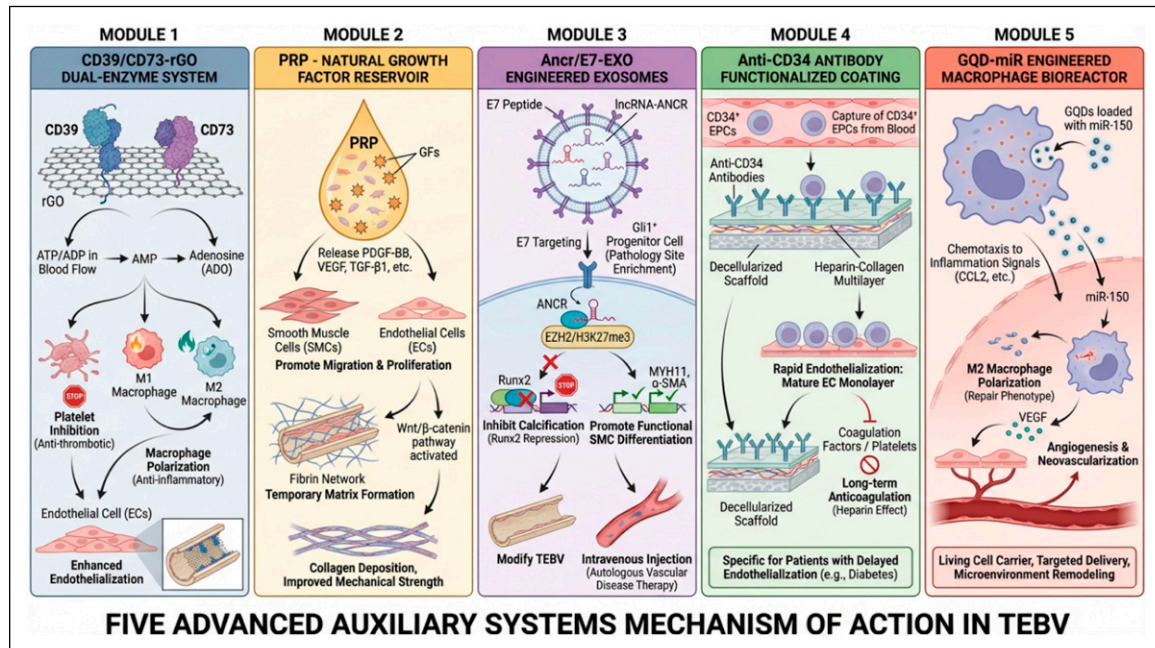


Figure 4. Representative auxiliary bioactive systems for antithrombosis, endothelialization, calcification control and immune regulation in tissue-engineered blood vessels. The figure summarizes stage-specific adjunctive concepts rather than clinically mature stand-alone platforms. Created by the authors based on published mechanistic concepts.^{79–91}

perfusion are not ornamental steps; they are often necessary to stabilize matrix architecture and to reveal failure modes before implantation. Newer reviews of small tissue engineered vessels therefore place greater emphasis on maturation workflows and functional testing, not just on scaffold formation.³¹ That shift is important because translation will depend increasingly on whether a graft has been stress-tested as a product rather than admired as an elegant construct (Tables 1–3).

3.2. Top-down matrix engineering: Mechanics versus bioactivity

Top-down matrix engineering approaches – including collagen/fibrin molding, densification and decellularization – prioritize immediate structural guidance. These methods are attractive because they can reproduce layered geometry and allow deliberate control over wall thickness, composition and cell placement.^{94,100–103}

Their weakness is that mechanical rescue often comes at a biological cost. Crosslinking or aggressive processing may improve handling and strength, yet can compromise cell viability, expose thrombogenic surfaces or reduce the very matrix cues that motivate the strategy.^{48–54,56,100–104} For this reason, top-down platforms are best viewed as controllable starting points that usually still require hemocompatible surface design and rigorous validation under flow.

This is one reason why validation has become as important as fabrication. As the literature grows, many grafts appear promising because they report burst pressure or short-term patency, yet far fewer are evaluated across coupled endpoints such as endothelial continuity, inflammatory phenotype, neointimal burden, calcification tendency and long-term compliance drift.^{23,31} A manuscript or platform that reports only one or two favorable metrics may still be informative mechanistically, but it is less convincing translationally. For small-diameter TEBVs, multi-parameter validation is no longer optional; it is the filter that distinguishes platform engineering from product engineering.

3.3. High-resolution biofabrication: Electrospinning and three-dimensional bioprinting

Electrospinning utilizes high-voltage electric fields to draw polymer solutions into micro- and nano-scale fibers, perfectly mimicking the fibrillar architecture of the native ECM.⁵⁴ By blending synthetic (PCL, PLA) and natural polymers (collagen, gelatin), researchers can precisely dictate porosity and degradation kinetics.^{55,56} The true power of electrospinning lies in hierarchical engineering: Ju et al. designed a dual-layer structure with an inner nanofiber layer to promote EC adhesion and an outer microfiber layer to facilitate SMC penetration. Similarly, McClure et al. achieved native-like arterial mechanics using a tri-layered PCL/elastin/collagen matrix, while Wu et al. utilized specialized collection techniques to achieve directed cellular

Table 1. Comprehensive comparison and decision matrix for tissue-engineered blood vessel seed-cell sources.

Cell type	Source	Key advantages	Key limitations	Vascular-Specific functions	References
Autologous Primary Vascular Cells	Patient's saphenous vein, internal mammary artery	No immune rejection Native elastic matching Low thrombogenicity	Donor site morbidity Poor cell quality in diseased patients Limited harvestable quantity	EC: Antithrombotic barrier SMC: Contractile function	61,62
Embryonic Stem Cells (ESCs)	Inner cell mass of blastocyst	Robust pluripotency Directed differentiation to EC/SMC Unlimited expansion potential Minimal ethical concerns	Ethical controversies Allogeneic immune rejection Teratoma formation risk	EC: In vivo perfusion restoration	65–67,69
Mesenchymal Stem Cells (MSCs)	Bone marrow, adipose tissue, umbilical cord	Immunomodulatory properties Relatively accessible	Limited differentiation plasticity Reduced function in aged/diseased donors High heterogeneity	SMC: Scaffold adhesion EC: Paracrine support	63,64,73
Induced Pluripotent Stem Cells (iPSCs)	Somatic cell reprogramming (fibroblasts, blood cells)	Patient-specific, no immune rejection Minimal ethical concerns Unlimited expansion potential	Epigenetic immaturity Potential tumorigenicity Unstable differentiation efficiency	SMC: Contractile phenotype maintenance EC: Functional vessel formation	70–73
Endothelial Progenitor Cells (EPCs)	Bone marrow mobilization, peripheral blood, umbilical cord blood	Rapid directed endothelialization Natural homing capacity Stable vascular network formation with MSCs	Source-dependent functional heterogeneity Limited EPC numbers in peripheral blood Long-term stability uncertain	EC: Rapid endothelialization Antithrombotic function	74–78

Note. EC = Endothelial Cells, SMC = Smooth Muscle Cells, TEV = Tissue-Engineered Blood Vessels.

alignment.¹¹⁹ The primary remaining hurdle for electrospinning is the inherently dense fiber packing, which often limits deep cellular infiltration.

To achieve scalable, off-the-shelf TEVVs with tunable properties, the field has pivoted toward advanced manufacturing. Electrospinning remains one of the most mature fabrication methods because it offers broad materials compatibility, architectural tunability and industrial familiarity. By adjusting fiber diameter, layering and collection strategy, researchers can influence cell alignment, porosity and regional mechanics.^{53–56,104–112,119}

3D bioprinting extends control one step further by allowing spatially programmed, multi-material constructs and patient-specific geometries. Representative examples include directly extruded vascular wall constructs, collagen-based coaxial biofabrication, rotary bioprinted small-diameter grafts and 3D-printed grafts with anti-thrombotic surface modification.^{113–118} The central barrier, however, remains the same: printability, mechanical stability and biological

Table 2. Evidence-aware summary of auxiliary bioactive systems for antithrombosis, endothelialization and immune regulation in tissue-engineered blood vessels.

Auxiliary system	Main functional target	Representative mechanism	Evidence level	Translational maturity	Main limitation	References
CD39/CD73-inspired purinergic regulation	Early thrombosis and inflammation	Hydrolysis of ATP/ADP toward adenosine-mediated signaling to reduce platelet activation and inflammatory amplification	Mechanistic and biomaterial-interface evidence	Early adjunctive concept	Enzyme stability, carrier biocompatibility and long-term performance under implantation conditions remain unresolved	79-81
Marine sulfated polysaccharides	Hemocompatibility and endothelial support	Sulfated glycans such as fucoidans, carrageenans and fucosylated chondroitin sulfates may mimic endothelial glycocalyx functions and modulate coagulation, inflammation and vascular cell behavior	Recent review-level synthesis and emerging scaffold-functionalization evidence	Emerging surface-functionalization strategy	Structural heterogeneity, sulfation pattern control, batch variability, anticoagulant potency and endothelial compatibility require standardization	98,99
Platelet-rich plasma	Matrix formation and early regenerative signaling	Autologous platelet-derived growth factors may support SMC migration, collagen deposition and local tissue regeneration	Preclinical and context-dependent adjunctive evidence	Variable autologous supplement	Dose dependence, preparation variability and risk of excessive initial response limit reproducibility	82,92,93
Ancr/E7-EXO engineered exosomes	Calcification and pathological lineage drift	E7-targeted exosomes carrying lncRNA-ANCR may redirect Gli1-positive progenitor cells by modulating EZH2/H3K27me3-Runx2 signaling	Small-animal and molecular targeting evidence	High-potential but early targeted adjunct	Scale-up, targeting efficiency, biodistribution and regulatory complexity remain major barriers	83-85
Anti-CD34 functionalized multilayers	Delayed endothelialization	Antibody-mediated capture of circulating progenitor cells combined with heparin-collagen layer-by-layer coating	Coating and in situ endothelialization evidence	Practical but patient-dependent surface strategy	Circulating EPC abundance and capture efficiency vary markedly across patients	86,87
GQD-miR macrophage bioreactor	Inflammatory microenvironment and angiogenesis	Macrophage-mediated delivery of miR-150 may promote M1-to-M2 polarization and pro-angiogenic signaling	Early proof-of-concept immune-programming evidence	Experimental living-cell delivery concept	In vivo fate tracking, cell-therapy regulation and safety control are unresolved	88

Note. EPCs = endothelial progenitor cells; GQD = graphene quantum dot; PRP = platelet-rich plasma; SMC = smooth muscle cell.

Table 3. Comparative analysis of advanced biofabrication technologies for tissue-engineered blood vessels.

Technology	Principle	Key advantages	Key limitations	References
Biological Self-Assembly (TESA)	Long-term cell sheet culture inducing ECM deposition → rolling	Highest biomimicry No synthetic material Burst pressure >2000 mmHg	Lengthy 8-month culture Difficult scalability	61,94,96
In Vivo Bioreactor	Polymer rod implantation → fibrous capsule formation → autologous graft harvest	Complete autologousization No in vitro culture	Patient-dependent performance Post-transplantation stenosis	28,30,95,97
Matrix Molding/Casting	Collagen/fibrin + cells → annular mold casting → plastic compression	Simple process High plasticity Burst pressure 1500-1800 mmHg	Chemical crosslinkers affect cell viability Shrinkage control difficult	94,100–103
Decellularized Matrices	Physical/chemical treatment removing cellular antigens → preserving native ECM architecture	Preserved native ECM cues Xenogeneic/allogeneic sources	Mechanical degradation Thrombogenic collagen exposure	48–56
Electrospinning	High-voltage electric field drawing polymer solutions → micro/nano-scale fiber deposition	Mature technology Wide material selection Industrial scalability	Dense fiber packing limits cell infiltration Solvent residue risk	53–56,104–112
3D Bioprinting	Computer-guided layer-by-layer deposition (extrusion, coaxial, photopolymerization)	Complex geometries Multi-material integration Personalization capability	Bioink paradox Resolution vs. viability trade-off	113–118

permissiveness rarely peak in the same bioink. The value of bioprinting therefore lies less in the promise of a universal solution than in its potential to integrate structural precision with application-specific graft design.

The same conclusion emerges from the broader vascular-engineering literature. Patterning, sacrificial templating and coaxial deposition are increasingly powerful for building structured conduits or vascularized tissues, but they do not by themselves guarantee arterial durability.³² Bioprinting is most compelling when it is used to answer a specific design problem – such as how to compartmentalize cell types, localize bioactivity or personalize geometry – rather than when it is presented as a universal manufacturing endpoint. For the foreseeable future, hybrid systems that combine printing with electrospinning, molding or post-print maturation may therefore be more realistic than any single-platform solution.^{31,32,118,120,121}

4. Molecular regulation of endothelialization and vascular remodeling

4.1. MicroRNA-mediated epigenetic regulation

MicroRNAs are better viewed as regulatory nodes than as isolated biomarkers. In TEBVs, they matter because early endothelialization, smooth-muscle phenotype and matrix remodeling are all sensitive to miRNA-controlled signaling networks. Representative angiomiRs such as miR-126 exemplify how small non-coding RNAs can bias endothelial survival and neovascularization, whereas other miRNAs restrain migration, proliferation or lineage stability.^{122–126}

This framing helps avoid a common mistake in review writing: treating molecular regulation as though it were a separate layer added after successful engineering. In reality, miRNA-mediated signaling only becomes clinically relevant if it can be coupled to a tractable delivery system or to a cell source with predictable behavior. The translational question is therefore not whether angiomiRs are important – they clearly are – but whether they can be used sparingly to reinforce specific product goals, such as faster endothelialization or suppression of osteogenic drift, without turning the graft into an uncontrollably complex gene-delivery device.^{85–87,123–126}

For vascular graft design, miRNA regulation is most relevant when it is coupled to a controllable delivery or presentation system. For example, miRNA-responsive endothelialization may be influenced by scaffold stiffness, surface chemistry, shear stress and local inflammatory cues. Thus, miRNAs should not be presented as isolated therapeutic molecules, but as molecular regulators whose effects depend on the material and hemodynamic context in which vascular cells are exposed.

4.2. Core signaling networks in endothelial fate and remodeling

Current pathway mapping suggests that these signals converge on a limited set of biologically meaningful axes, including VEGF/VEGFR, PI3K/AKT, MAPK/ERK, EGFR, BMPR 2 and TGF-beta-associated programs.^{122,127–132} For TEBV research, the practical implication is not that every graft should become a nucleic-acid delivery device, but that molecular regulation helps explain why ostensibly similar constructs remodel differently after implantation.

Accordingly, miRNA knowledge is most valuable when used sparingly and mechanistically: to support endothelialization, suppress osteogenic drift, or inform the design of exosome- and nanoparticle-based adjuncts.¹³³ Figure 5 summarizes this regulatory layer as a complement to – rather than a replacement for – scaffold and cell engineering.

The same restraint applies to angiogenesis more broadly. Vascularization is indispensable, but excessive or poorly organized angiogenesis can also destabilize neotissue, increase permeability and intensify inflammatory remodeling. The most useful molecular programs are therefore those that coordinate endothelial survival, mural support and matrix deposition rather than simply maximizing vessel sprouting. This is another reason why molecular regulation should be interpreted through the lens of indication and product logic: what is desirable in a regenerative research model may not be desirable in a conduit that must remain mechanically quiet and surgically reliable for years.^{23,32}

A more useful translational framework is to map molecular pathways onto engineering failure modes. VEGF/VEGFR and PI3K/AKT signaling are most relevant to endothelial survival and barrier formation; MAPK/ERK and TGF-beta-associated programs may influence proliferation, matrix deposition and maladaptive remodeling; osteogenic regulators such as Runx2 become critical when calcification risk is high. This mapping helps identify which molecular pathway should be modulated in a given graft design, rather than treating angiogenesis as a uniformly desirable endpoint.

5. Indication-specific translation and validation

The preceding sections indicate that translational criteria differ across coronary bypass, distal peripheral reconstruction, hemodialysis access and urgent vascular trauma. The following section therefore compares these indications according to their

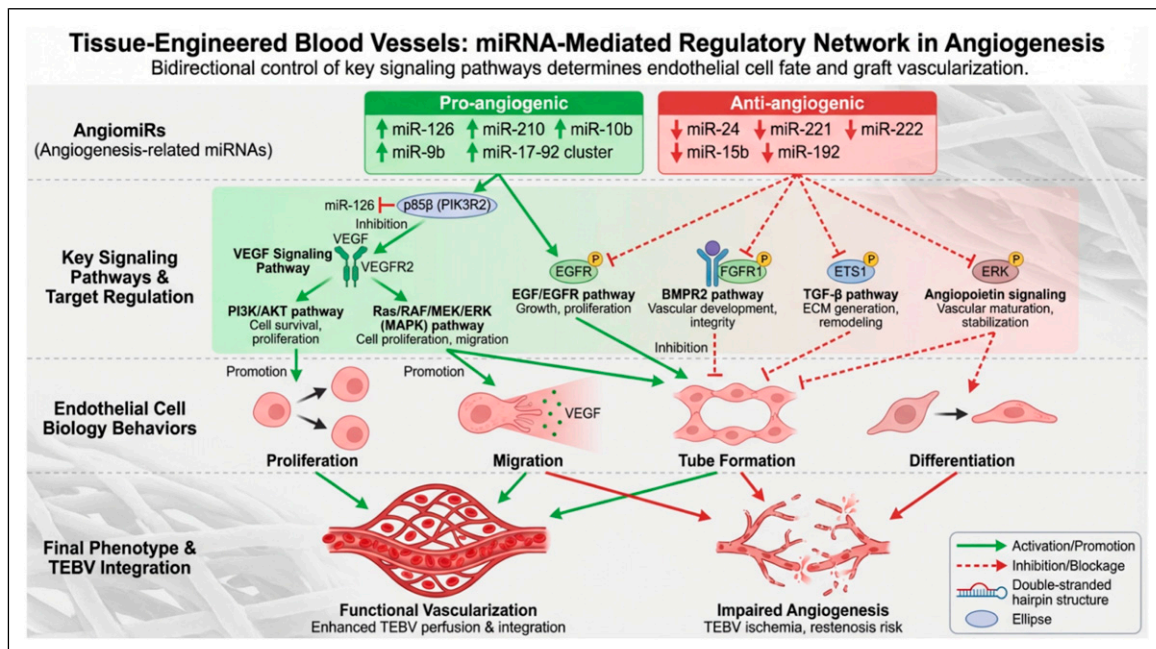


Figure 5. Simplified microRNA-regulated angiogenic network relevant to tissue-engineered blood vessel integration. The scheme is intended as a mechanistic overlay for scaffold and cell design, not as an exhaustive pathway atlas. Created by the authors based on published pathway evidence.^{122–133}

dominant failure modes, evidence maturity and product-development constraints. This structure is intended to clarify how specific design features should be evaluated rather than to restate the broader argument for indication-matched graft development.

5.1. Indication-specific design logic

Coronary bypass sets the strictest biological standard. The target diameter is small, competitive flow may be present, distal runoff varies, and even modest thrombotic or hyperplastic responses can compromise graft patency. In this setting, rapid endothelialization and luminal hemocompatibility are not optional refinements but baseline requirements. A graft that is mechanically adequate yet slow to endothelialize is unlikely to compete with internal thoracic or radial artery conduits. Consequently, coronary-directed TEBV strategies should be judged primarily by how reliably they establish a stable antithrombotic surface under arterial shear rather than by how biomimetic they appear in static culture.^{5,8,23,24,26}

Below-knee or distal peripheral reconstruction imposes a different emphasis. These grafts face challenging outflow beds, diffuse atherosclerotic disease and often a heavily inflammatory systemic environment, particularly in patients with diabetes or chronic limb-threatening ischemia. Here, the problem is not only thrombosis but maladaptive remodeling in compromised host tissue. Composite scaffolds, matrix-informed bioactivity and selective immune control may therefore be more relevant than highly personalized cellular approaches. A distal peripheral platform must tolerate poor biological terrain and still maintain a usable lumen; this raises the value of durability, controllable degradation and inflammation-aware design.^{5,8,23,24,29}

Hemodialysis access occupies an intermediate position in which infection resistance, surgical handling and repetitive puncture tolerance matter alongside patency. Clinical and translational experience with biotubes, tissue-engineered access grafts and host-repopulating constructs suggests that this indication may be one of the most realistic proving grounds for advanced vascular substitutes.^{28,30,134} The necessary conduit length is substantial, the clinical need is recurrent, and modest biological variability may be acceptable if the product remains available off the shelf or can be prepared in a semi-standardized manner. For this reason, access applications may continue to reward robust decellularized or acellular designs before they reward highly elaborate cell-laden constructs.

Urgent vascular trauma, by contrast, is the clearest case for an immediately available acellular product.²⁹ There is no time for autologous manufacturing, and the comparator is not a perfect native artery but the imperfect real-world alternative of synthetic grafts or absent usable vein. The recent regulatory and clinical trajectory of the acellular tissue engineered vessel therefore matters beyond trauma alone: it demonstrates that a human-derived, off-the-shelf conduit can meet a sufficiently specific clinical need to support structured clinical validation and early real-world trauma use.^{25,27} Trauma does not prove that all TEBVs are ready for routine use, but it does prove that indication-matched product logic can succeed.

5.2. Biofabrication, validation and the end of trial-and-error

Manufacturing remains a rate-limiting step. Even when a construct performs well in a short-term animal model, translation depends on reproducible wall architecture, sterilization compatibility, storage strategy, suture handling, fatigue resistance and testing under physiologically relevant flow. Recent reviews emphasize that validation criteria for small-diameter grafts must extend beyond burst pressure to include compliance, kink resistance, thrombogenicity, endothelial coverage and remodeling kinetics.¹³⁵

The practical implication is that trial-and-error design is becoming increasingly inefficient. As graft systems become more layered and biologically active, each added degree of freedom multiplies the number of possible failure modes. Computational fluid dynamics, fluid-structure interaction modeling and vascular microphysiological systems are therefore most useful when they reduce this design space before expensive large-animal or clinical studies are attempted.^{23,135,136} Used properly, these tools do not replace implantation studies; they make those studies more informative by filtering out architectures that are mechanically or hemodynamically implausible from the outset.

Computational modeling and microphysiological systems are valuable here because they can narrow the design space before large-animal or clinical testing. Organ-on-a-chip platforms, fluid-structure interaction modeling and related in vitro systems help link geometry and material behavior to local wall stress, nutrient transport and endothelial response under flow.^{136,137} These tools are best viewed as accelerators of disciplined iteration, not substitutes for in vivo evidence.

Because the evidence base ranges from in vitro assays and small-animal implantation to early human observational studies, clinically relevant endpoints should be interpreted according to evidence maturity. Patency is particularly difficult to compare across studies because definitions, imaging methods, follow-up intervals and target indications differ substantially. Table 4 summarizes representative studies in which model type, follow-up duration, sample size and patency-related outcomes can be linked to translational interpretation.

Table 4. Representative translational and clinically relevant evidence for tissue-engineered vascular grafts.

Platform or study type	Model or population	Sample size/study scale	Follow-up	Patency or functional endpoint	Translational interpretation	References
Scaffold-free, cell-assembled tissue-engineered blood vessel	Adult human-cell-derived grafts evaluated in long-term animal models	Preclinical animal evaluation	Long-term animal follow-up reported in the original study	Patency and mechanical function were assessed in animal arterial bypass settings, together with burst pressure and vessel handling properties	Demonstrates biological fidelity and mechanical feasibility, but manufacturing time and patient-specific production limit broad clinical deployment	61
In-body tissue-engineered autologous Biotube	First human use for hemodialysis access	Early first-in-human clinical report	Three-year follow-up	Longitudinal access function and patency-related follow-up were reported, but the evidence remains early and not comparative	Supports feasibility of in vivo bioreactor-derived autologous grafts, especially for access applications, while highlighting patient dependence and scalability constraints	28
Acellular tissue-engineered vessel for extremity vascular injury	Subset of vascular injury participants from two open-label, single-arm, nonrandomized clinical trials	51 patients in the prospective civilian trial subset and 16 patients in the war-zone observational subset	Primary endpoint at day 30; longer follow-up ongoing	Day-30 primary patency was reported as 84.3% and 93.8% in the two study subsets; secondary patency was reported as 90.2% and 93.8%, respectively	Provides the strongest current clinical signal for an off-the-shelf acellular conduit in urgent vascular trauma, but the evidence is nonrandomized and indication-specific	25
Acellular tissue-engineered vessel in a combat setting	Real-world combat-related vascular injury cohort	Retrospective observational multicenter experience	Follow-up extended beyond the early perioperative period in the original report	Patency, limb salvage, infection and safety outcomes were evaluated in a real-world trauma context	Supports feasibility under challenging trauma conditions, but should not be interpreted as randomized evidence of superiority over autologous vein or conventional synthetic grafts	27
Marine sulfated polysaccharide-functionalized vascular biomaterials	Emerging hemocompatible and endothelialization-oriented scaffold functionalization	Review-level synthesis and early scaffold functionalization evidence	Mostly preclinical or materials-level evidence	Patency has not yet been established as a mature clinical endpoint for this strategy	Highlights a new antithrombotic and endothelial-support surface chemistry, but translation depends on standardization and in vivo validation	92,93

(continued)

Table 4. (continued)

Platform or study type	Model or population	Sample size/study scale	Follow-up	Patency or functional endpoint	Translational interpretation	References
Electrospun or printed synthetic/natural hybrid scaffolds	In vitro studies and small-animal vascular graft models	Mostly preclinical scale	Short- to mid-term animal follow-up depending on study	Patency and remodeling are variably reported; direct comparison is limited by heterogeneity in diameter, animal species, anticoagulation and implantation site	Supports architectural tunability, but clinical credibility requires standardized patency, endothelial coverage, inflammation and compliance reporting	53–56,104–119,138

5.3. Clinical development and regulatory relevance

Clinical translation is no longer hypothetical. Acellular or host-repopulating vascular conduits have progressed from feasibility reports to structured clinical evaluation, and recent analyses of the Humacyte acellular tissue-engineered vessel platform show how off-the-shelf bioengineered vessels may fill specific niches such as vascular trauma when autologous vein is not feasible.^{20,25,27,98} The broader lesson is that early clinical translation may favor products with clear indications, standardized manufacture and a manageable biologic burden.

However, the available ATEV evidence should be interpreted with appropriate caution. The JAMA Surgery vascular injury analysis and the Military Medicine combat-setting study provide important clinical and real-world signals, but they are observational or nonrandomized analyses in trauma populations rather than randomized comparisons against standard-of-care reconstruction.^{25,27} These studies therefore support feasibility and indication-specific product logic, but they do not establish general superiority over autologous vein or conventional synthetic grafts across all vascular indications.

From a product-development perspective, the value of these studies lies in their clinical specificity. They link conduit performance to endpoints such as patency, limb salvage, infection and multicenter usability in settings where autologous vein may be unavailable or unsuitable. This makes them more informative for urgent vascular trauma than for coronary bypass, distal peripheral reconstruction or hemodialysis access, where comparator standards and failure mechanisms differ substantially.

5.4. Future tissue-engineering design strategies

Future progress is most likely to come from technologies that reinforce, rather than destabilize, translational discipline.

First, smart biodegradable elastomers and responsive composite scaffolds may improve the coordination of mechanics and remodeling, particularly when they are used to tune degradation or local release rather than to create unnecessarily complex materials.^{22,24,99} Their value will depend on whether they solve a bounded problem – such as delayed endothelialization or late calcification – without undermining sterilization, storage or reproducibility.

Second, hybrid fabrication strategies that combine hierarchical micro/nano-architecture with controlled luminal bioactivity may offer a realistic middle ground between biologically rich but slow-to-manufacture constructs and simple but underperforming synthetics.^{31,32} In this framework, 3D printing, electrospinning and molded ECM-based layers are best regarded as interoperable tools rather than competing ideologies. Hybrid systems are especially attractive when they can localize biological complexity to the lumen while keeping the supporting wall structure straightforward and scalable.

Third, cell and immune engineering – including exosome-guided lineage control, selective endothelial recruitment and macrophage programming – should be judged by whether they solve the dominant failure mode of a defined indication, not by conceptual novelty alone.^{85–90} The nearer-term opportunity is selective biological assistance, whereas fully autonomous or gene-circuit-driven grafts remain longer-range ambitions. In other words, the field should prefer targeted gains in clinical credibility over maximal conceptual ambition, because the latter often expands complexity faster than it expands utility.

6. Conclusions and recommendations

TEBV research has moved beyond the question of whether a biologically active vascular substitute can be built. The more important question is which combination of scaffold design, cell strategy, bioactive regulation and manufacturing route can survive the sequence of demands imposed by implantation, blood flow and host remodeling.

The evidence reviewed here supports a restrained conclusion: clinically credible TEBVs should be developed as indication-specific products rather than as universal vessel substitutes. Future studies should therefore define the intended clinical task before selecting material complexity, cellular strategy or adjunctive bioactivity. Success should be judged not only by early patency, but also by endothelial continuity, remodeling stability, infection resistance, mechanical durability, puncture tolerance where relevant, and manufacturability.

This product-oriented view also implies a higher standard for reporting. Studies should provide coupled mechanical, biological and functional endpoints, including degradation kinetics, compliance, suture retention, thrombogenicity, endothelial coverage, inflammatory phenotype, calcification tendency and patency-related follow-up whenever implantation models are used. Without such evidence, an elegant construct may remain scientifically interesting but translationally ambiguous.

The most durable impact of recent work may therefore be conceptual. Tissue-engineered vascular grafting is moving away from the search for a single ideal vessel and toward a portfolio model in which materials, cells and adjunctive biology are

assembled according to indication, evidence maturity and regulatory realism. If pursued rigorously, this shift offers a clearer route from laboratory prototypes to clinically credible vascular substitutes.

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Consent to participate

Informed consent was not required because this article is a narrative review and does not involve human participants.

Author contributions

Junjie Chen, Jiayang He and Dujiang Yang contributed equally to this work. Junjie Chen, Chunshui He and Wei Zeng conceived the review framework. Junjie Chen, Jiayang He, Dujiang Yang, Yuhan He and Junhao Zhang drafted and revised the manuscript. Chunshui He and Wei Zeng supervised the work and critically revised the manuscript. All authors read and approved the final manuscript.

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During preparation of this work, the authors used ChatGPT for language polishing, formatting assistance and editorial organization. After using this tool, the authors reviewed and edited all outputs as needed and take full responsibility for the final content of the article.

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